

OR Central Exercise

Column Generation

12 June 2026

Question 1

There are 20m and 17m boards, and an order of 30 type A (6m), 15 type B (7m), and 40 type C (3m) boards. It costs X euros to use Xm boards for $X \in \{17, 20\}$. We want to minimize cost.

part a)

For CG, an optimal cutting pattern must be generated w.r.t. a given utility vector $u = (u_A, u_B, u_C) \in \mathbb{R}^3$ (can be the shadow prices) and a given length $l \in \{17, 20\}$. Formulate an IP that solves the problem "find an optimal cut w.r.t. u and l ".

Let y_T for $T \in \{A, B, C\}$ be the number of type T obtained from the new cutting pattern.

$$\max_y \quad u^T y - c_l = \sum_T \underline{u_T y_T} - \underline{c_l}$$

s.t. feasible cutting pattern $\rightarrow 6y_A + 7y_B + 3y_C \leq c_l$

$y_A, y_B, y_C \in \mathbb{N}_0$

we write this model
for each $l \in \{17, 20\}$

variable for
finding best a,
(cutting pattern)

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part b)

Start with 3 cutting patterns where 1 of each type is cut from a board of length $17m$. Run one iteration of the CG algorithm with this basis.

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Initial basis: (x_1, x_2, x_3)

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$x_B = b = \begin{pmatrix} 30 \\ 15 \\ 40 \end{pmatrix}$$

$$c_B = \begin{pmatrix} 17 \\ 17 \\ 17 \end{pmatrix}$$

$$(17 \ 17 \ 17) y - c_y \quad \max_{y \in \mathbb{N}_0} c_B^T B^{-1} y - c_y$$

$$= 17y_A + 17y_B + 17y_C - c_y \quad \text{s.t.} \quad \sum_T l_T y_T \leq l_y \rightarrow 6y_A + 7y_B + 3y_C \leq l_y$$

$$c_y = l_y \in \{17, 20\}$$

Solution for $17m$

$$(0, 0, 5) \Rightarrow obj = 68$$

New basis: (x_1, x_2, x_4)

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 6 \end{pmatrix}$$

$$c_B = \begin{pmatrix} 17 \\ 17 \\ 20 \end{pmatrix}$$

Solution for $20m$

$$\text{new pattern } x_4 \leftarrow (0, 0, 6) \Rightarrow obj = 82$$

Question 2

m people with $w_i \in \mathbb{N}$ kg luggages. Each wants to travel in the same car with their luggage. Car capacity is $W \in \mathbb{N}$ kg. S denotes the set of all quarrelsome pairs $\{i, j\}$. Minimize the number of cars.

$A \in \mathbb{R}^{m \times n}$ describes permissible filling patterns, $c = (1, \dots, 1)^T \in \mathbb{R}^n$ and $b = (1, \dots, 1)^T \in \mathbb{R}^m$. At least one person must travel in each car.

$$\begin{aligned} \min, & \sum_{i=1}^n c_i x_i \\ \text{s.t.} & Ax = b \\ & x \in \{0, 1\}^n. \end{aligned}$$

part a)

Formulize valid filling patterns, " $a = (a_1, \dots, a_m)^T \in \{0, 1\}^m$ " through linear (in)equality constraints.

y_i : variable

- At least one person in the car: $\sum_{i=1}^m y_i \geq 1$
- Total weight is not exceeded: $\sum_{i=1}^m y_i w_i \leq W$
- Quarrelsome people not in the same car: $y_i + y_j \leq 1 \quad \forall \{i, j\} \in S.$

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$$\begin{aligned} \min, & \sum_{i=1}^n c_i x_i \\ \text{s.t.} & Ax = b \\ & x \in \{0, 1\}^n. \end{aligned}$$

part b)

Letting $w_i = 10$ for all $i \in \{1, \dots, m\}$, $W = 40$, and for an arbitrary $m \in \mathbb{N}$, specify the number n , the columns of A for the two sets $S_1 = \emptyset$ and $S_2 = \{\{i, j\} : i \neq j\}$.

- $S_1: \binom{m}{4} + \binom{m}{3} + \binom{m}{2} + \binom{m}{1}$
- $S_2: \quad \quad \quad \binom{m}{1} = m$

total #
of valid
seating
plans

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$$\begin{aligned} \min, & \sum_{i=1}^n c_i x_i \\ \text{s.t.} & Ax = b \\ & x \in \{0, 1\}^n. \end{aligned}$$

part c)

Under **what condition on the number $|S|$** of quarrelsome pairs is it particularly advisable to use the Column Generation Algorithm? Use the results from subtask b) to justify your answer.

Number of seating plans for part b):

use CG when $|S|$ is small.

For S_1 , it was $\sim m^4$ when $|S_1| = 0$.

decreases $\nwarrow \leftarrow$ increase

For S_2 , it was m when $|S_2| = \binom{m}{2}$.

Question 2

part d)

Letting $m = 4$, $w_i = 10$, $W = 40$, and $S = \{\{1, 2\}, \{2, 3\}\}$, perform one iteration of the CG algorithm, starting with the basis in which each person travels alone.

Initial basis: (x_1, x_2, x_3, x_4)

x_i : person i travels alone

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I \quad c_B = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\max_{y \in \mathbb{N}_0} c_B^T B^{-1} y - c_y \rightarrow (1111) I y - c_y = \sum_{i=1}^4 y_i - 1$$

$$\sum_{i=1}^4 10 \cdot y_i \leq 40 \leftarrow \text{s.t. } \sum_i w_i y_i \leq W$$

$$\sum_i y_i \geq 1 \rightarrow y_1 + y_2 + y_3 + y_4 \geq 1$$

$$y_1 + y_2 \leq 1, y_2 + y_3 \leq 1 \leftarrow y_i + y_j \leq 1 \quad \forall \{i, j\} \in S.$$

Optimal solution:

$$y = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$$

New column: x_5 , new var.
 $\dots + x_5 = 1$
 $\dots + \dots = 4$
 $+ x_5 = 1$
 $\dots + x_5 = 1$

Exiting variable:

can be x_1, x_3
or x_4